

MH2100 Calculus III

Midterm I (Solutions)

AY 2019/2020, Semester 1

Quantitative Research Society @NTU

September 2019

Overview

- **Coverage:** parametric curves and tangent lines; limits, continuity and partial derivatives; tangent planes, the Chain Rule, implicit differentiation, and critical points.
- **Format:** three questions, 100 marks in total, 90 minutes (official paper: September 2019).
- **Historical permitted material:** the September 2019 paper permitted one two-sided A4 piece of paper with handwritten notes. Consult current course materials for present assessment rules.
- **Source:** MH2100 Midterm I, AY2019/2020 Semester 1. Questions are transcribed from the official paper; solutions and marking schemes are unofficial QRS@NTU material.

Instructions (verbatim)

1. DO NOT TURN OVER PAPER UNTIL INSTRUCTED.
2. This midterm paper contains THREE (3) questions.
3. Answer ALL questions. The marks for each question are indicated at the beginning of each question.
4. Candidates can write anywhere on this midterm paper.
5. ONE two-sided A4 piece of paper with handwritten notes is permitted.

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Question 1**(30 Marks)****QUESTION 1.**

- (i) Find parametric equations to represent the line segment from $(2, 3)$ to $(5, 5)$.
- (ii) Write a vector equation for a parametrisation of the tangent line at the point when $t = \pi$ to the curve with parametric equations

$$x = \cos t, \quad y = \sin t, \quad z = t.$$

- (iii) Write the parametrisation of a curve that has tangent vector $\langle e^t, 2t, \sin t \rangle$.
- (iv) Find the point(s) on the following curve where the tangent is horizontal.

$$x = t + 4, \quad y = t^3 - 3t + 7.$$

- (v) Find the directional derivative of the function $f(x, y, z) = xy^2 - xz^2$ in the direction of the vector $\langle 1, 2, 3 \rangle$.

Solution**Method 1: Direct computation**

- (i) A segment from $\mathbf{p} = \langle 2, 3 \rangle$ to $\mathbf{q} = \langle 5, 5 \rangle$ is

$$\mathbf{r}(t) = \mathbf{p} + t(\mathbf{q} - \mathbf{p}) = \langle 2, 3 \rangle + t\langle 3, 2 \rangle, \quad 0 \leq t \leq 1.$$

Thus

$$\boxed{x = 2 + 3t, \quad y = 3 + 2t, \quad 0 \leq t \leq 1.}$$

- (ii) Let $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$. Then

$$\mathbf{r}(\pi) = \langle -1, 0, \pi \rangle, \quad \mathbf{r}'(\pi) = \langle 0, -1, 1 \rangle.$$

Hence a parametrisation of the tangent line is

$$\boxed{\mathbf{l}(s) = \langle -1, 0, \pi \rangle + s\langle 0, -1, 1 \rangle, \quad s \in \mathbb{R}.}$$

- (iii) Integrate the tangent vector componentwise. One suitable curve is

$$\boxed{\mathbf{r}(t) = \langle e^t, t^2, -\cos t \rangle,}$$

because $\mathbf{r}'(t) = \langle e^t, 2t, \sin t \rangle$. Arbitrary constant components may be added.

- (iv) Since $dx/dt = 1$, a horizontal tangent occurs exactly when

$$\frac{dy}{dt} = 3t^2 - 3 = 0,$$

so $t = \pm 1$. The corresponding points are

$$\boxed{(5, 5) \text{ and } (3, 9).}$$

- (v) The unit vector in the stated direction is

$$\mathbf{u} = \frac{\langle 1, 2, 3 \rangle}{\sqrt{14}}.$$

Also,

$$\text{grad } f(x, y, z) = \langle y^2 - z^2, 2xy, -2xz \rangle.$$

Therefore

$$D_{\mathbf{u}}f(x, y, z) = \frac{y^2 - z^2 + 4xy - 6xz}{\sqrt{14}}.$$

Unofficial QRS marking scheme (30 marks)

- (i) **(6 marks)** 3: correct displacement vector; 2: correct affine parametrisation; 1: restricts the parameter to $0 \leq t \leq 1$.
- (ii) **(6 marks)** 2: finds the point at $t = \pi$; 2: finds the tangent vector; 2: gives a line through that point with that direction.
- (iii) **(6 marks)** 4: integrates all three components correctly; 2: verifies the derivative or allows an arbitrary constant vector.
- (iv) **(6 marks)** 2: differentiates x and y ; 2: solves for both parameter values and checks $x' \neq 0$; 2: gives both points.
- (v) **(6 marks)** 2: normalizes the direction vector; 2: computes the gradient; 2: takes the correct dot product.

Question 2**(35 Marks)****QUESTION 2.**

- (i) Evaluate each of the following limits or show that the limit does not exist. Justify your answers.

(a)

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y + 5}{\sin(xy) + 1}$$

(b)

$$\lim_{(x,y) \rightarrow (0,0)} \frac{7xy^3}{xy^3 - (x - y)^2}$$

(c)

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy^3}{x^2 + y^6}$$

- (ii) Define

$$f(x, y) = \begin{cases} y \cos\left(\frac{xy}{x^2 + y^2}\right); & \text{if } x^2 + y^2 \neq 0, \\ 0; & \text{if } x^2 + y^2 = 0. \end{cases}$$

- (a) Is f continuous at $(0, 0)$? Justify your answer.
 (b) Find $f_x(0, 0)$ and $f_y(0, 0)$.
 (c) Is f differentiable at $(0, 0)$? Justify your answer.

Solution**Method 1: Direct limits and definitions**

- (i) (a) The denominator tends to 1, so the quotient is continuous at $(0, 0)$. Hence

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y + 5}{\sin(xy) + 1} = 5.$$

- (b) Along $x = 0$, the quotient equals 0 for $y \neq 0$. Along $x = y$, it equals 7 for $x \neq 0$. Since these restricted limits differ,

the limit does not exist.

- (c) Along $x = 0$, the quotient equals 0. Along $x = y^3$, it is

$$\frac{y^6}{y^6 + y^6} = \frac{1}{2} \quad (y \neq 0).$$

The restricted limits differ, so

the limit does not exist.

- (ii) (a) Since $|\cos u| \leq 1$,

$$|f(x, y)| \leq |y| \leq \sqrt{x^2 + y^2} \longrightarrow 0 = f(0, 0).$$

Thus

f is continuous at $(0, 0)$.

(b) By the definitions of the partial derivatives,

$$f_x(0,0) = \lim_{h \rightarrow 0} \frac{f(h,0) - f(0,0)}{h} = 0,$$

while

$$f_y(0,0) = \lim_{k \rightarrow 0} \frac{f(0,k) - f(0,0)}{k} = \lim_{k \rightarrow 0} \frac{k}{k} = 1.$$

Therefore

$$\boxed{f_x(0,0) = 0, \quad f_y(0,0) = 1.}$$

(c) If f were differentiable, its differential would be $L(h,k) = k$. Along $(h,k) = (t,t)$, with $t \neq 0$,

$$f(t,t) = t \cos\left(\frac{1}{2}\right),$$

and hence

$$\frac{f(t,t) - L(t,t)}{\sqrt{t^2 + t^2}} = \frac{\cos(1/2) - 1}{\sqrt{2}} \frac{t}{|t|}.$$

This does not tend to 0 as $t \rightarrow 0$. Therefore

$$\boxed{f \text{ is not differentiable at } (0,0).}$$

Unofficial QRS marking scheme (35 marks)

(i) (15 marks)

- (a) (5 marks) 2: observes a nonzero limiting denominator; 2: evaluates the numerator and denominator; 1: gives 5.
- (b) (5 marks) 2: computes the $x = 0$ restricted limit; 2: computes the $x = y$ restricted limit; 1: concludes nonexistence.
- (c) (5 marks) 2: computes the $x = 0$ restricted limit; 2: computes the $x = y^3$ restricted limit; 1: concludes nonexistence.

(ii) (20 marks)

- (a) (5 marks) 3: establishes a valid bound tending to zero; 2: concludes continuity.
- (b) (5 marks) 2: correct difference quotient for f_x ; 2: correct difference quotient for f_y ; 1: states both values.
- (c) (10 marks) 2: identifies the only possible linear map $L(h,k) = k$; 4: evaluates a valid path; 2: normalizes the remainder; 2: concludes non-differentiability.

Question 3**(35 Marks)****QUESTION 3.**

- (i) Find an equation for the tangent plane to the surface $z = 5x^2y - 7xy^2$ at the point $(3, 2, 6)$.
- (ii) Find an equation for the tangent plane to the surface given by the equation $x^2y - yz + xz^2 = 17$ at the point $(1, 2, 5)$.

- (iii) Use the Chain Rule to find $\frac{\partial f}{\partial u}$ and $\frac{\partial f}{\partial v}$ for

$$f(x, y, z) = 2xy^2z^4, \quad x = 2uv, \quad y = u^2, \quad z = \sin v.$$

- (iv) Find $\frac{\partial z}{\partial y}$ where z is defined implicitly by

$$xe^{4y} + 4y^2z^3 + ze^{8x} = 0.$$

- (v) Let

$$f(x, y) = x^3 - 3x^2y + 3xy^2 - 3x - 9y.$$

Find and classify all critical points of f . (For each critical point, determine whether or not it is a local max/min or a saddle point.)

Solution**Method 1: Direct computation**

- (i) Let $g(x, y) = 5x^2y - 7xy^2$. Then

$$g_x = 10xy - 7y^2, \quad g_y = 5x^2 - 14xy,$$

so $g_x(3, 2) = 32$ and $g_y(3, 2) = -39$. The tangent plane is

$$z - 6 = 32(x - 3) - 39(y - 2), \quad \boxed{32x - 39y - z = 12.}$$

- (ii) Put $F(x, y, z) = x^2y - yz + xz^2 - 17$. Then

$$\text{grad } F(x, y, z) = \langle 2xy + z^2, x^2 - z, -y + 2xz \rangle,$$

and $\text{grad } F(1, 2, 5) = \langle 29, -4, 8 \rangle$. Thus the tangent plane is

$$29(x - 1) - 4(y - 2) + 8(z - 5) = 0, \quad \boxed{29x - 4y + 8z = 61.}$$

- (iii) By the Chain Rule,

$$f_u = f_x x_u + f_y y_u + f_z z_u, \quad f_v = f_x x_v + f_y y_v + f_z z_v.$$

Here

$$f_x = 2y^2z^4, \quad f_y = 4xyz^4, \quad f_z = 8xy^2z^3,$$

and

$$x_u = 2v, \quad y_u = 2u, \quad z_u = 0, \quad x_v = 2u, \quad y_v = 0, \quad z_v = \cos v.$$

Substitution gives

$$\boxed{\frac{\partial f}{\partial u} = 20u^4v \sin^4 v,}$$

$$\boxed{\frac{\partial f}{\partial v} = 4u^5 \sin^4 v + 16u^5v \sin^3 v \cos v.}$$

- (iv) Let $F(x, y, z) = xe^{4y} + 4y^2z^3 + ze^{8x}$. Holding x constant and differentiating with respect to y gives

$$4xe^{4y} + 8yz^3 + (12y^2z^2 + e^{8x})\frac{\partial z}{\partial y} = 0.$$

Provided $12y^2z^2 + e^{8x} \neq 0$,

$$\frac{\partial z}{\partial y} = -\frac{4xe^{4y} + 8yz^3}{12y^2z^2 + e^{8x}}.$$

- (v) The first partial derivatives are

$$f_x = 3x^2 - 6xy + 3y^2 - 3 = 3((x - y)^2 - 1),$$

$$f_y = -3x^2 + 6xy - 9 = -3((x - y)^2 - y^2 + 3).$$

Putting $d = x - y$, the equation $f_x = 0$ gives $d^2 = 1$. The equation $f_y = 0$ then gives $y^2 = d^2 + 3 = 4$. Hence the four critical points are

$$(-3, -2), \quad (-1, -2), \quad (1, 2), \quad (3, 2).$$

The Hessian and its determinant are

$$H = \begin{pmatrix} 6(x - y) & -6(x - y) \\ -6(x - y) & 6x \end{pmatrix}, \quad \det H = 36y(x - y).$$

At $(-3, -2)$ the determinant is 72 and $f_{xx} = -6$, giving a strict local maximum. At $(3, 2)$ the determinant is 72 and $f_{xx} = 6$, giving a strict local minimum. At the other two points the determinant is -72 , giving saddle points. Thus

$$\begin{array}{l} (-3, -2) : \text{strict local maximum,} \quad (3, 2) : \text{strict local minimum,} \\ (-1, -2) \text{ and } (1, 2) : \text{saddle points.} \end{array}$$

Method 2: Structural route for Question 3(v)

The invertible linear coordinates $(d, y) = (x - y, y)$ separate the objective:

$$f = (d^3 - 3d) + (y^3 - 12y).$$

The first summand has a strict local maximum at $d = -1$ and minimum at $d = 1$; the second has a strict local maximum at $y = -2$ and minimum at $y = 2$. Combining two maxima or two minima gives the corresponding local extremum. Combining opposite types gives a saddle, because varying one coordinate increases the objective while varying the other decreases it. The change of coordinates preserves these local classifications and gives the same four points above.

Unofficial QRS marking scheme (35 marks)

- (i) **(7 marks)** 3: correct partial derivatives; 2: evaluates both slopes at $(3, 2)$; 2: gives the correct plane through $(3, 2, 6)$.
- (ii) **(7 marks)** 3: defines F and computes its gradient; 2: evaluates the normal at $(1, 2, 5)$; 2: gives the correct plane.
- (iii) **(7 marks)** 3: writes the correct Chain Rule; 2: differentiates the substitutions; 1: correct f_u ; 1: correct f_v .

- (iv) **(6 marks)** 3: differentiates the implicit equation correctly; 2: collects the z_y terms; 1: solves for z_y with its nonzero-denominator condition.
- (v) **(8 marks)** Either method earns full credit:
- Method 1: 2 for f_x, f_y ; 2 for all four critical points; 2 for the Hessian discriminants; 2 for the maximum, minimum, and two saddles.
 - Method 2: 2 for the invertible change $d = x - y$ and separation; 2 for the one-variable maxima and minima; 2 for mapping all four points; 2 for the resulting classifications.